Sanghack Lee, Ph.D.

Contact Information

Graduate School of Data Science
Seoul National University
Seoul, Republic of Korea

+82-2-880-9776
sanghack@snu.ac.kr
sanghack.lee@gmail.com

Research Interests

Causal and Counterfactual inference (identification and estimation) and causal discovery in diverse settings including tabular, relational, and (non-)stationary time series. Sequential decision making problems from the aspect of causality. Practical causal modeling for common sense. Incorporating concepts from human behavior (e.g., irrationality) into a causal framework.

Affiliation

Graduate School of Data Science (associate professor)
Artificial Intelligence Institute (adjunct professor)

Employment

Seoul National University, Seoul
Associate Professor, Graduate School of Data Science — **Mar. 2025–present**
Assistant Professor, Graduate School of Data Science — **Mar. 2021–Feb. 2025**

Columbia University, New York
Associate Research Scientist, Computer Science — **Jul. 2019–Feb. 2021**

Investigating ranges of causality problems from classical causal inference to its application to sequential decision making.

Purdue University, West Lafayette
Postdoctoral Research Associate, Computer Science — **Apr. 2018–Jun. 2019**
Advisor: Prof. Elias Bareinboim

University Service

SNU College, Seoul National University, Seoul
Director, Center for Innovative Digital Education — **Oct. 2026–Sep. 2028**
Head of Applied Studies — **Oct. 2026–Sep. 2028**

Education

Pennsylvania State University, University Park
Ph.D., College of Information Sciences and Technology — **2018**
Advisor: Prof. Vasant G. Honavar

Sogang University, Seoul, Republic of Korea
MS., Computer Science and Engineering — **2006**
BE., Computer Science and Engineering, *Cum Laude* — **2004**

Publications

✉ for corresponding authors (since 2021).

Sangyeon Cho, Minyoung Cho, Jungsoo Kim, Sujeong Oh, **Sanghack Lee**✉ (2026) Decomposing Conformal Uncertainty: Calibration- and Instance-Driven Feature Attribution. In *Advances in Neural Information Processing Systems (NeurIPS 2026)* (to appear)

{Yesong Choe, Yeahoon Kwon}, Minjung Kho, Seunggeun Lee, **Sanghack Lee**✉ (2026) Event-Time Regression Discontinuity for Biomedical Data: LATT Estimation and Covariate Distribution Tests. In *American Journal of Epidemiology* (accepted)

Jonghwan Kim, **Sanghack Lee**✉ (2026) Don't Test What You Can Deduce: Causal Discovery with Logical Inference. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)* Oral

Yesong Choe, {Yeahoon Kwon, Min Woo Park}, **Sanghack Lee**✉ (2026) Canonical Domain Reduction for Partial Counterfactual Identification. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)*

Teahan Kim, Minyoung Cho, **Sanghack Lee**✉ (2026) Beyond Bounds: Quantifying the Probability of Counterfactual Fairness. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)*

Min Woo Park, **Sanghack Lee**✉ (2026) On Transportability for Structural Causal Bandits. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)*

Yoonseok Yeom, Jonghwan Kim, Taehui Yun, Juhyun Lyu, Jung-Hee Kim, Sangmin Lee, Jinseok Yang, Hyemin Jung, Woohyung Lim, **Sanghack Lee**✉ (2026) Estimating Interventional Outcomes over Time with Causal Normalizing Flow. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)*

Kwonho Kim, Heejeong Nam, Inwoo Hwang, **Sanghack Lee**✉ (2026) On Causal Representation Learning with Internal Auxiliaries. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)*

Min Woo Park, Taehui Yun, YoungIn Jang, Yoonseok Yeom, Jonghwan Kim, Jiyeon Kang, Songseong Kim, Hyemin Jung, Sangmin Lee, Jongseong Jang, **Sanghack Lee**✉ (2026) Breaking Bad: Component-Wise Parent Deletion for Score-Based Causal Discovery. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)*

Min Woo Park, **Sanghack Lee**✉ (2026) Counterfactual Structural Causal Bandits. In *Proceedings of the 14th International Conference on Learning Representations (ICLR 2026)*

Hyeonji Kim, Sujeong Oh, **Sanghack Lee**✉ (2026) Mitigating Length Bias in RLHF through a Causal Lens. In *Proceedings of the 40th AAAI Conference on Artificial Intelligence (AAAI 2026)*

Junha Ham, Deokgyu Kim, Doeun Kim, Serjin Kim, **Sanghack Lee**✉ (2026) Causal Foundation Models Perform Better without Post-treatment Variables. In *ICML 2026 Workshop on Foundation Models for Structured Data (FMSD) (non-archival)*

Junha Ham, Deokgyu Kim, Doeun Kim, Serjin Kim, **Sanghack Lee**✉ (2026) A Structural View of Query Misspecification in Causal Foundation Models. In *ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM) (non-archival)*

Sangyeon Cho, Minyoung Cho, Jungsoo Kim, Sujeong Oh, **Sanghack Lee**✉ (2026) A Tale of Two Uncertainties: Global–Local Attribution for Conformal Prediction. In *ICML 2026 Workshop on Epistemic Intelligence in Machine Learning (EIML) (non-archival)*

Min Woo Park, Andy Arditi, Elias Bareinboim✉, **Sanghack Lee**✉ (2025) Structural Causal Bandits under Markov Equivalence. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*

Yeahoon Kwon, Yesong Choe, Soungmin Park, Neil Dhir✉, **Sanghack Lee**✉ (2025) Non-Stationary Structural Causal Bandits. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*

Junhyung Ahn, **Sanghack Lee**✉ (2025) On Predicting Post-Click Conversion Rate via Counterfactual Inference. In *Proceedings of the 25th IEEE International Conference on Data Mining (ICDM 2025)* Best Paper Award Finalist

Dong Kyu Cho, Inwoo Hwang, **Sanghack Lee**✉ (2025) PEER pressure: Model-to-Model Regularization for Single Source Domain Generalization. In *Proceedings of the 41st IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2025)*

Jungsoo Kim, Kwonho Kim, Inwoo Hwang, **Sanghack Lee**✉ (2025) Instrumental Variable Representation Learning under Confounded Covariates. In *NeurIPS 2025 Workshop: CauScien — Uncovering Causality in Science (non-archival)*

Chanhui Lee*, Juhyeon Kim*, YongJun Jeong, Yeom Yoon Seok, Juhyun Lyu, Jung-Hee Kim, Sangmin Lee, Sangjun Han, Hyeokjun Choe, Soyeon Park, Woohyung Lim, Kyunghoon Bae,

Sungbin Lim✉, **Sanghack Lee**✉ (2024) On Incorporating Prior Knowledge Extracted from Pretrained Language Models into Causal Discovery In *First Workshop on Causality and Large Model at NeurIPS 2024 (non-archival)*

Sujin Jeon, Inwoo Hwang, **Sanghack Lee**✉, Byoung-Tak Zhang✉ (2024) Locality-aware Concept Bottleneck Model In *UniReps: 2nd Edition of the Workshop on Unifying Representations in Neural Models at NeurIPS 2024 (non-archival)*

Yonghan Jung✉, Min Woo Park, **Sanghack Lee**✉ (2024) Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*

Ryan Carey, **Sanghack Lee**, Robin J. Evans (2024) Toward a Complete Criterion for Value of Information in Insoluble Decision Problems. In *Transactions on Machine Learning Research 2024*.

Inwoo Hwang, Yunhyeok Kwak, Suhyung Choi, Byoung-Tak Zhang✉, **Sanghack Lee**✉ (2024) Fine-Grained Causal Dynamics Learning with Quantization for Improving Robustness in Reinforcement Learning. In *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*

{Inwoo Hwang, Yesong Choe}, Yeahoon Kwon, **Sanghack Lee**✉ (2024) On Positivity Condition for Causal Inference. In *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*

Jonghwan Kim, Inwoo Hwang, **Sanghack Lee**✉ (2024) Causal Discovery with Deductive Reasoning: One Less Problem. In *Proceedings of the 40th Conference on Uncertainty in Artificial Intelligence (UAI 2024)*

Yunhyeok Kwak, Inwoo Hwang, Dooyoung Kim, **Sanghack Lee**✉, Byoung-Tak Zhang✉ (2024) Efficient Monte Carlo Tree Search via On-the-Fly State-Conditioned Action Abstraction. In *Proceedings of the 40th Conference on Uncertainty in Artificial Intelligence (UAI 2024)*

Soheun Yi, **Sanghack Lee**✉ (2024) Filter, Rank, and Prune: Learning Linear Cyclic Gaussian Graphical Models. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS 2024)*

Inwoo Hwang, Yunhyeok Kwak, Suhyung Choi, Byoung-Tak Zhang, **Sanghack Lee**✉ (2023) Quantized Local Independence Discovery for Fine-Grained Causal Dynamics Learning in Reinforcement Learning. In *GenPlan 2023: Seventh Workshop on Generalization in Planning at NeurIPS 2023 (non-archival)*

Dong Kyu Cho, **Sanghack Lee**✉ (2023) Learning to ignore: Single Source Domain Generalization via Oracle Regularization. In *Causal Representation Learning Workshop at NeurIPS 2023 (non-archival)*

Inwoo Hwang, Yunhyeok Kwak, Suhyung Choi, Byoung-Tak Zhang, **Sanghack Lee**✉ (2023) Causal Dynamics Learning with Quantized Local Independence Discovery. In *Second Workshop on Spurious Correlations, Invariance and Stability at ICML 2023 (non-archival)*

Inwoo Hwang, Yunhyeok Kwak, Yeon-Ji Song, Byoung-Tak Zhang, **Sanghack Lee**✉ (2023) On Discovery of Local Independence over Continuous Variables via Neural Contextual Decomposition. In *Proceedings of the 2nd Conference on Causal Learning and Reasoning (CLeaR 2023)*

Juhyeon Kim, Yesong Choe, **Sanghack Lee**✉ (2022). Detecting Causality by Data Augmentation via Part-of-Speech tagging. In *CASE: The 5th Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text at EMNLP (2022)*

Juan D. Correa, **Sanghack Lee** and Elias Bareinboim (2022). Counterfactual Transportability: A Formal Approach. In *Proceedings of the 39th International Conference on Machine Learning (ICML 2022)*

Inwoo Hwang, Byoung-Tak Zhang, **Sanghack Lee**✉ (2021). Partition-based Local Independence Discovery. Workshop "Causal Inference Challenges in Sequential Decision Making: Bridging Theory and Practice" at NeurIPS 2021.

Juan D. Correa, **Sanghack Lee**, and Elias Bareinboim (2021). Nested Counterfactual Identification from Arbitrary Surrogate Experiments. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*

Sanghack Lee and Elias Bareinboim (2021). Causal Identification with Matrix Equations. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*

— before affiliated with Seoul National University —

Sanghack Lee and Elias Bareinboim (2020). Characterizing Optimal Mixed Policies: Where to Intervene and What to Observe. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*

Sanghack Lee and Elias Bareinboim (2020). Causal Effect Identifiability under Partial Observability. In *Proceedings of the 37th International Conference on Machine Learning (ICML 2020)*

Sanghack Lee, Juan D. Correa, and Elias Bareinboim (2020). General Transportability — Synthesizing Observations and Experiments from Heterogeneous Domains. In *Proceedings of the Thirty-fourth AAAI Conference on Artificial Intelligence (AAAI 2020)*

Sanghack Lee, Juan D. Correa, and Elias Bareinboim (2020). Identifiability from a Combination of Observations and Experiments. In *Proceedings of the Thirty-fourth AAAI Conference on Artificial Intelligence (AAAI 2020)*

Sanghack Lee, Juan D. Correa, and Elias Bareinboim (2019). General Identifiability with Arbitrary Surrogate Experiments. In *Proceedings of Thirty-fifth Conference on Uncertainty in Artificial Intelligence (UAI 2019)* Best Paper Award

Sanghack Lee and Vasant Honavar (2019). Towards Robust Relational Causal Discovery. In *Proceedings of Thirty-fifth Conference on Uncertainty in Artificial Intelligence (UAI 2019)*

Aria Khademi, **Sanghack Lee**, David Foley, Vasant Honavar (2019). Fairness in Algorithmic Decision Making: An Excursion Through the Lens of Causality. In *Proceedings of 2019 International Conference on World-Wide Web (WWW 2019)*

Sanghack Lee and Elias Bareinboim (2019). Structural Causal Bandits with Non-manipulable Variables. In *Proceedings of Thirty-third AAAI Conference on Artificial Intelligence (AAAI 2019)*

Sanghack Lee and Elias Bareinboim (2018). Structural Causal Bandits: Where to Intervene?. In *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*

Sanghack Lee and Vasant Honavar (2017). Self-Discrepancy Conditional Independence Test. In *Proceedings of Thirty-third Conference on Uncertainty in Artificial Intelligence (UAI 2017)*

Sanghack Lee and Vasant Honavar (2017). A Kernel Conditional Independence Test for Relational Data. In *Proceedings of Thirty-third Conference on Uncertainty in Artificial Intelligence (UAI 2017)*

Sanghack Lee and Vasant Honavar (2016). A Characterization of Markov Equivalence Classes of Relational Causal Models under Path Semantics. In *Proceedings of Thirty-second Conference on Uncertainty in Artificial Intelligence (UAI 2016)*

Kyungsik Han, **Sanghack Lee**, Jin Yea Jang, Yong Jung, and Dongwon Lee (2016). "Teens are from Mars, Adults are from Venus": Analyzing and Predicting Age Groups with Behavioral Characteristics in Instagram. In *Proceedings of Eighth International ACM Web Science Conference 2016 (WebSci 2016)*

Sanghack Lee and Vasant Honavar (2016). On Learning Causal Models for Relational Data. In *Proceedings of Thirtieth Conference on Artificial Intelligence (AAAI 2016)*

Sanghack Lee and Vasant Honavar (2015). Lifted Representation of Relational Causal Models Revisited: Implications for Reasoning and Structure Learning. In *Proceedings of the UAI 2015 Workshop on Advances in Causal Inference co-located with the 31st Conference on Uncertainty in Artificial Intelligence (UAI 2015)*

{Elias Bareinboim, **Sanghack Lee**}, Vasant Honavar, and Judea Pearl (2013). Transportability from Multiple Environments with Limited Experiments. In *Advances in Neural Information Processing Systems 26 (NeurIPS 2013)*

Sanghack Lee and Vasant Honavar (2013). m-Transportability: Transportability of a Causal Effect from Multiple Environments. In *Proceedings of the Twenty-seventh Conference on Artificial Intelligence (AAAI 2013)*

Sanghack Lee and Vasant Honavar (2013). Causal Transportability of Experiments on Controllable Subsets of Variables: z-Transportability. In *Proceedings of the Twenty-ninth Conference on Uncertainty in Artificial Intelligence (UAI 2013)*

{Harris Lin, **Sanghack Lee**, Ngot Bui} and Vasant Honavar (2013). Learning Classifiers from Distributional Data. In *IEEE Second International Congress on Big Data*

Sanghack Lee, Jihoon Yang and Sungyong Park (2006). A New Polynomial Time Algorithm for Bayesian Network Structure Learning. In *Advanced Data Mining and Applications (ADMA 2006)*

Sanghack Lee, Jihoon Yang and Sung-Yong Park (2004). Discovery of Hidden Similarity on Collaborative Filtering to Overcome Sparsity Problem. In *Discovery Science 2004 (DS 2004)*

Talks, Tutorials, Posters

- Pacific Causal Inference Conference (July, 2026, scheduled), invited talk
- Seminar, Samsung Advanced Institute for Health Sciences & Technology (June, 2026, scheduled), invited talk
- Pukyong National University (May, 2026, scheduled), invited tutorial
- Data Science ABC for Social Scientists (Aug, 2025), tutorial
- Samsung DS (Feb, 2025), invited tutorial
- Pukyong National University (Nov, 2024), invited tutorial
- DS+ Tutorial Series, Causal Inference in the Age of AI, Korea University (Aug, 2024), invited tutorial
- Recent Advances in Causal Inference, Korean AI Association (Aug, 2024), invited talk
- KUBIG Korea University (Dec, 2023), invited talk
- BK21 Big Data Science Seminar, Yonsei University (Nov, 2023), invited talk
- Digital Convergence Branch, The Korean Neurosurgical Society (Nov, 2023), invited talk
- Recent Trends in Causal Inference, Korean Society of Artificial Intelligence in Medicine (Oct, 2023), invited talk
- On the Role of Causal Inference in Artificial Intelligence, SNU AI.MED Talks Series (Project Group for Education and Research in Medical AI) (Sep, 2023), invited talk
- Tutorial on Causal Inference, Korean AI Association (July, 2023), invited tutorial
- AI Seminar, Samsung Electronics, Device Solution (June, 2023), invited talk
- Tutorial on Causal Inference, Korean AI Association (Jan, 2023), invited tutorial
- Seminar at Integrated Major in Innovative Medical Science, College of Medicine, Seoul National University (Nov, 2022), invited talk

- Artificial Intelligence Graduate School (AIGS), Ulsan National Institute of Science and Technology (UNIST) (Nov, 2022), invited talk
- 2022 Korea Summer Workshop on Causal Inference Program / Korean AI Association (Jul, 2022), invited talk
- Korean Society of Epidemiology (Jul, 2022), invited tutorial
- The Korean Statistical Society (Jun, 2022), invited talk
- Statistics Department Seminar, Seoul National University (Jun, 2022), invited talk
- Seminar, Samsung Advanced Institute for Health Sciences & Technology (May, 2022), invited talk
- GSDS Seminar, Seoul National University (Mar, 2022), invited talk
- Institute of Economic Research, Seoul National University (Dec, 2021), invited talk
- Preventive Medicine, College of Medicine, Yonsei University (Dec, 2021), invited talk
- Amazon Research at Tübingen Germany (Aug, 2021), invited talk
- Applied BigData Engineering Seminar at Sogang University (Jun, 2021), invited talk
- NeurIPS'2020 (Virtual), poster
- Summer AI Seminar Series at POSTECH (Aug, 2020) (Pohang, South Korea), invited talk
- ICML'2020 (Virtual), poster
- AAAI'2020 (New York, NY), talk
- AAAI'2020 (New York, NY), invited talk
- UAI'2019 (Tel Aviv, Israel), talk (*)
- IJCAI'2019 (Macau), "Causal Reinforcement Learning", tutorial (*)
- ISysE Seminar at KAIST (Apr, 2019) (Daejeon, South Korea), invited talk
- AAAI'2019 (Hawaii), talk
- NeurIPS'2018 (Montreal, Canada), poster
- Causality workshop at UAI'2017 (Sydney, Australia), talk
- UAI'2017 (Sydney, Australia), two posters
- UAI'2016 (Jersey City, NJ), talk
- AAAI'2016 (Phoenix, AZ), talk
- Causality workshop at UAI'2015 (Amsterdam, Netherlands), poster
- AAAI'2013 (Bellevue, WA), talk
- UAI'2013 (Bellevue, WA), poster

* someone else substituted for me.

Professional Service

Board Member

- Korean Artificial Intelligence Association (2023, 2024, 2025)
- KIISE Artificial Intelligence Society (2024, 2025)

Program Committee / Reviewers (* as an external reviewer)

- 2027 AAAI (Senior Program Committee), AISTATS, ICLR (Area Chair), ICML (Senior Area Chair)

- 2026 AAAI, AISTATS, ICML (Area Chair), UAI, CLeaR, NeurIPS (Top Area Chair, Workshop Proposal Reviewer)
- 2025 AAAI, KDD (Outstanding Reviewer), ICLR, AISTATS, CLeaR, ICML (Area Chair), NeurIPS (Top Area Chair), JMLR
- 2024 ICLR, CLeaR, JMLR, ICML, UAI, NeurIPS (Area Chair), ARR June, ECAI, CI4TS@UAI
- 2023 UAI, JMLR, NeurIPS (Top Reviewer),
- 2022 ICLR (Highlighted Reviewer), AAAI, AISTATS, CLeaR, ICML, Journal of Causal Inference (JCI), UAI (Top Reviewer), NeurIPS*
- 2021 ICLR, AAAI, AISTATS, ICML, UAI, NeurIPS, Journal of Artificial Intelligence Research (JAIR), NeurIPS 2021 Workshop on 'Causal Inference & Machine Learning: Why now?'
- 2020 NeurIPS, UAI, ICML (Top Reviewer Award), AAAI, AISTATS, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), Journal of Artificial Intelligence (AIJ), Journal of Causal Inference (JCI), NeurIPS Workshop on Causal Discovery and Causality-Inspired Machine Learning (CDML, Area Chair), ACM-IMS Foundations of Data Science Conference (FODS)*, Statistical Science*
- 2019 NeurIPS (Best Reviewer Award), Journal of Machine Learning Research (JMLR), WHY conference (AAAI Spring Symposium), ICML*, IJCAI*
- 2018 NeurIPS*
- 2017 Causality Workshop at UAI
- 2016 ACM CHI Conference on Human Factors in Computing Systems
- 2014 ACM Transactions on Intelligent Systems and Technology (TIST)

Industrial Experience

Senior Engineer at **Diquest, inc.**, Seoul, South Korea — **Feb. 2006 to Jun. 2009**

Development and maintenance of an enterprise search engine

Research Experience

Associate Research Scientist, Columbia University — **Jul. 2019–Feb. 2021**

Post-doctoral Research Associate, Purdue University — **2018–Jun. 2019**

Research Assistant, Pennsylvania State University — **2013–2014, 2015–2018**

Research Assistant, Iowa State University — **2011–2013**

Research Assistant, Sogang University — **2005**

Teaching Experience

Lecture, Seoul National University

- Causal Inference for Artificial Intelligence (Fall 2024)
- Causal Inference for Data Science (Fall 2021, Spring 2023, Spring 2024, Fall 2024)
- Machine Learning and Deep Learning for Data Science II (Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024 co-taught w/ Prof. Taesup Kim)
- Project for Data Science (Spring 2023)
- Big Data and Knowledge-based System II (Fall 2022, Spring 2023, co-taught w/ Prof. Wen-Syan Li)
- Introduction to Artificial Intelligence (Spring 2022)
- Introduction to Big Data* (Spring 2022, planned, co-taught by GSDS professors)
- Computing Foundations for Data Science (Winter Bootcamp 2022, 300+ students, co-taught with Prof. Joonseok Lee)

- Principles and Application of Data Science (Fall 2021, together with other professors)
- Foundations of Data Science (Summer 2021, co-taught by GSDS professors)

Guest Lecture, Purdue University

Structural Causal Bandits (Advanced Machine Learning, Spring 2019), Counterfactual Bandits (Advanced Machine Learning, Spring 2019)

Guest Lecture, Columbia University

Structural Causal Bandits (Causal Inference II, Spring 2021 and Spring 2022)

Graduate Teaching Assistant

(Pennsylvania State University): Discrete Mathematics, Principles of Artificial Intelligence.

(Iowa State University): Design and Analysis of Algorithms (2 times), Principles of Artificial Intelligence, Machine Learning, Object-Oriented Analysis and Design.

(Sogang University): Java Language Programming, Personal Computer Laboratory I, Discrete Structures.

Advisee

Ph.D. students

Taehui Yun	(Sep 2026–)
Bogeun Kim	(Sep 2026–)
Sangyeon Cho	(Sep 2026–)
Kim, Jung Soo	(Sep 2024–)
Kim, Jonghwan	(Sep 2024–)
Lim, Byeonghui	(July 2024–)
Yeo Dong Youn	(Jan 2024–)
Park, Min Woo	(Jan 2023–)
Choe, Yesong	(July 2021–)
Kwon, Yeahoon	(July 2021–)

Master students

Woongi Yoon	(Sep 2026–)
Kangsan Kim	(Sep 2025–)
Youngin Jang	(Sep 2025–)
Hyunwoo Kim	(Apr 2025–)
GyeongChan Han	(Mar 2025–)
Juyong Sung	(Mar 2025–)

Alumni

Hwang, Inwoo (Ph.D.)	(Mar 2021–Feb 2025)
Min Young Cho	(Sep 2024–Aug 2026)
Sujeong Oh	(July 2024–Feb 2026)
Daehui Park	(July 2024–Feb 2026)
Yoonseok Yeom	(July 2024–Feb 2026)
Eunseo Lee	(July 2024–Feb 2026)
Sumin Cho	(July 2024–Feb 2026)
Kwon, Oh Yoon	(Jun 2023–Aug 2025)
Park, Hyunwoo	(Jun 2023–Aug 2025)
Jihae Chung	(Jan 2024–Aug 2025)

Hyeonji Kim	(Jan 2024–Aug 2025)
Jeoung, Jaeho	(Jan 2023–Aug 2025)
Kim, Kwon Ho	(Jun 2023–Feb 2025)
Choi, Jin A	(Jun 2023–Feb 2025)
Park, Soungmin	(Jun 2023–Feb 2025)
Choi, Heejin	(Jan 2023–Feb 2025)
Kim, Taehan	(July 2021–Feb 2025)
Song, Kyung A	(Jan 2023–Aug 2024)
Yim, Dahhee	(July 2022–Aug 2024)
Park, Jeongsup	(July 2022–Feb 2024)
Moon, Jeong Ha	(July 2022–Feb 2024)
Soheun Yi	(Mar 2022–Aug 2023, undergraduate)
Cho, Dong Kyu	(July 2021–Jun 2023)
Chung, Chaeyoung	(July 2021–Jun 2023)
Kim, Juhyeon	(July 2021–Feb 2023)
Kang, Jewon	(July 2021–Feb 2023)

Grants

Advancing Credit Decision Making, AFINIT (PI, 2025–2026)
Scalable Causal Discovery, LG AI Research (PI, 2025–2026)
Deep Generative Models for Causal Reasoning, LG AI Research (PI, 2024–2025)
Causal Discovery for Time Series Data Guided by Data-driven Causal Knowledge, LG AI Research (PI, 2023–2024)
Causal Machine Learning, NRF, (PI, 2023–2027)
Self-Directed Artificial Intelligence, IITP, (co-PI, 2022–2026)
Center for Optimizing Hyperscale AI Models and Platforms, NRF, (co-PI, 2023–2030)
Analyzing Associations between Metabolomics Big Data and Adverse Drug Reactions, MFDS, (co-PI, 2023–2025)
Algorithmic Characterization of Matrix-based Causal Inference, SNU, (PI, 2021–2024)

References

Prof. Vasant Honavar
Professor
Information Sciences and Technology,
Pennsylvania State University, USA
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Prof. Elias Bareinboim
Associate Professor
Computer Science,
Columbia University, USA
eb@cs.columbia.edu

Prof. Jihoon Yang
Professor
Computer Science and Engineering,
Sogang University, South Korea
yangjh@sogang.ac.kr

Prof. Jin Tian
Professor
Machine Learning,
Mohamed bin Zayed University of Artificial Intelligence, UAE
jin.tian@mbzuai.ac.ae

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